

Customer Churn Prediction using Machine Learning

Student Name	Dhanavath Navadeep	Internship	CodeAlpha
Project Type	Machine Learning Classification	Dataset Target	Churn (Binary Status)
GitHub Repo	github.com/DhanavathNavadeep/Task_1_ML_Classification		

Project Objective

The objective of this project is to develop an enterprise-grade machine learning classification pipeline to predict customer churn dynamics accurately. The operational boundary spans end-to-end processing data pipelines, handling structural features, validating multi-algorithm architectures, and validating model alignment parameters through systematic comparisons.

Tools and Technologies

Python

Pandas

NumPy

Scikit-learn

Matplotlib

Jupyter Notebook

GitHub

Methodology Architecture

To ensure stability and maximize predictive feature correlation, the architecture was segmented into distinct pipeline steps:

1. Data Engineering & Preprocessing

- Rigorous handling and cleaning of missing dataset elements.
- Robust Label Encoding of dense categorical string vectors.
- Targeted feature selection algorithms to limit variance overhead.

2. Training & Validation Setup

- Structured Train/Test split isolation configurations.
- Multi-fold Stratified Cross-Validation checks.
- Parallel training runs to compare diverse classifiers.

Algorithms Implemented

- **Logistic Regression:** Utilized as a linear classification baseline mapping generalized odds tracking matrices.
- **Random Forest Classifier:** Deployed as an ensemble tree structure to capture non-linear parameter interactions and minimize ensemble structural variance.

Evaluation Framework

Model optimization and comparative scoring runs were calculated based on the following mathematical performance criteria:

- **Accuracy & Precision:** Overall predictive certainty tracking and reduction of false positive markers.
- **Recall & F1-Score:** Sensitivity threshold balancing to detect maximum true churn vectors.
- **ROC-AUC Score:** Area under the receiver operating characteristic curve to assert model discriminatory power.

Results & Comparative Performance

Pipeline Validation Complete:

- Both individual pipelines successfully initialized, iterated, and exported.
- Comprehensive verification metrics compiled alongside graphical evaluation plots including the **Confusion Matrix, ROC Curve, and Feature Importance Diagrams.**
- The ensemble model successfully isolated higher relative performance criteria and was tagged for serialization.

Conclusion

This classification project successfully showcases an industrial-grade engineering implementation covering the complete machine learning model development cycle. By executing thorough ingestion workflows, cross-validating multi-architecture models, and mapping extensive statistical boundaries, the framework drastically improves understanding of supervised predictive engines and provides a reproducible deployment foundation for enterprise risk mitigation.